Task: Year 10 Term 1 Week 4 test prep

1. Expand and simplify:
$4(y+3)-5y+6$

By the order of operations we want to expand the brackets first.
To expand the brackets we can use the distributive law: $A(B+C)=AB+AC$

To use the distributive law we multiply each of the terms inside the brackets by 4.

= $4y+12-5y+6$

Simplify the expression by collecting the like terms.

$4y$ and $-5y$ are like terms. To combine them we add the coefficients.
12 and 6 are like terms. Add them together.

= $-y+18$

2. Expand and simplify:
$c(c+7)+5(c+6)$

By the order of operations we want to expand the brackets first.
To expand the brackets we can use the distributive law: $A(B+C)=AB+AC$

Multiply each of the terms in the first set of brackets by c.
Then multiply each of the terms in the second set of brackets by 5.

= $c^2+7c+5c+30$

Simplify the expression by collecting the like terms.

We can combine the terms in c by adding the coefficients.

= $c^2+12c+30$

3. Expand and simplify the following:
$(m-2)(m+9)$

To expand $(m-2)(m+9)$, think of $(m+9)$ as a single quantity.
We can then apply the distributive law $A(B+C)=AB+AC$ by multiplying $(m+9)$ by both m and -2.

We can rewrite the distributive law as $(x+a)(x+b)=x(x+b)+a(x+b)$.
We can use this law to multiply m with $(m+9)$ and -2 with $(m+9)$.

$(m-2)(m+9)$ = $m(m+9)-2(m+9)$

We want to expand both of $m(m+9)$ and $-2(m+9)$ using the distributive law $A(B+C)=AB+AC$.

Multiply m with each term in $(m+9)$, and multiply -2 with each term in $(m+9)$

$(m-2)(m+9)$ = $m^2+9m-2m-18$

Add the like terms.

In this case, $9m$ and $-2m$ are like terms that can be added.

$(m-2)(m+9)$ = $m^2+7m-18$

4. Expand the following using binomial expansion:
$\left(x+3\right)\left(x+10\right)$

Use the distributive law to expand the brackets

Multiply x with each term in $x+10$ and 3 with each term in $x+10$.

$\left(x+3\right)\left(x+10\right)$ = $\left(x\right)^2+10x+3x+30$

Collect like terms and simplify

$10x+3x=13x$

$\left(x+3\right)\left(x+10\right)$ = $x^2+13x+30$

5. Factorise the following expression by taking out the highest common factor:
$14pqr-5pst$

Find the highest common factor (HCF) of the two terms and use $AB-AC=A\left(B-C\right)$ to factorise.

Since the highest numerical factor common to both terms is 1, the common algebraic component will be the HCF.

$14pqr-5pst$ = $p\left(14qr-5st\right)$

6. Factorise the following expression by taking out the highest common factor:
$31x-x^2$

Find the highest common factor (HCF) of the two terms and use $AB-AC=A(B-C)$ to factorise.

Since the highest numerical factor common to both terms is 1, the common algebraic component will be the HCF.

$31x-x^2$ = $x(31-x)$

7. Express $\frac{4x}{4}$ in simplest form.

Cancel out the common numerical factor from the numerator and denominator.

$4x$ and 4 have a highest common factor of 4.

$\frac{4x}{4}$ = x

8. Simplify the expression $\frac{2t^3}{3t^3}$. ------

To simplify a fraction, we cancel out any common factors in the numerator and denominator. What is a common factor of the numerator and denominator?

The coefficients 2 and 3 have no common factor. However, t^3 is a factor of both the numerator and the denominator.
What do we get when we divide both the numerator and denominator by t^3?

$\frac{2t^3}{3t^3} = \frac{2}{3}$

9. Rewrite $\frac{-3}{k}$ with $5k$ as the denominator. ------

$$\frac{-3}{k} = \frac{\square}{5k}$$

We need to compare the denominator on the left with the one on the right to determine what factor is missing. We can then multiply both the numerator and the denominator of $\frac{-3}{k}$ by that factor to get an equivalent fraction with denominator $5k$.

What factor do we have to multiply the denominator k by in order for it to be equal to the denominator $5k$?

The denominator $5k$ is the denominator k multiplied by a factor of 5. So, we can multiply the numerator and denominator of $\frac{-3}{k}$ by 5 to get an equivalent fraction with denominator of $5k$.

$\frac{-3}{k}$ = $\frac{-3}{k} \times \frac{5}{5}$

Multiply the numerators and multiply the denominators.

$\frac{A}{B} \times \frac{C}{D} = \frac{A \times C}{B \times D}$.

$\frac{-3}{k}$ = $\frac{-15}{5k}$

10. Simplify $\frac{3x}{6}+\frac{4x}{6}$ ------

Since both fractions have a common denominator of 6, we can express them as a single fraction.

$\frac{3x}{6}+\frac{4x}{6}$ = $\frac{3x+4x}{6}$

Simplify the numerator by collecting like terms.

Since the numerator has two terms containing a single power of x, $3x+4x$ can be simplified to $7x$.

$\frac{3x}{6}+\frac{4x}{6}$ = $\frac{7x}{6}$

11. Simplify $\frac{3x}{35}-\frac{2x}{15}$. ------

The lowest common multiple of 35 and 15 is 105 . So rewrite both fractions with a denominator of 105

To do this multiply the first term by $\frac{3}{3}$ and the second term by $\frac{7}{7}$

$= \frac{3x \times 3}{35 \times 3} - \frac{2x \times 7}{15 \times 7}$

Evaluate each numerator and denominator

$= \frac{9x}{105} - \frac{14x}{105}$

Since we now have a common denominator, express as a single fraction by combining numerators

$= \frac{9x - 14x}{105}$

Simplify the terms in the numerator

$9x - 14x = -5x$

$= \frac{-5x}{105}$

Simplify the fraction by dividing the numerator and denominator by their highest common factor

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Divide each term by 5

$= \frac{-x}{21}$

12. Simplify $\frac{c}{4} \times \frac{d}{3}$.

Multiply numerator by numerator and denominator by denominator.

$\frac{c}{4} \times \frac{d}{3}$ = $\frac{c \times d}{4 \times 3}$

Evaluate the products in the numerator and denominator.

$c \times d = cd$

$\frac{c}{4} \times \frac{d}{3}$ = $\frac{cd}{12}$

13. Simplify the following:
$\frac{10}{9y} \times \frac{11}{9y}$

To express this as a single fraction, we can multiply the numerators and multiply the denominators.

The numerator becomes 10×11.
The denominator becomes $9y \times 9y$. What is the result of multiplying y by y?

= $\frac{110}{81y^2}$

14. Simplify $\frac{-7x}{8} \div \frac{9x}{10}$.

Dividing by a fraction is equal to multiplying by its reciprocal.

The reciprocal of $\frac{9x}{10}$ can be found by swapping its numerator and denominator.

$= \frac{-7x}{8} \times \frac{10}{9x}$

Cancel out the common algebraic factor.

$= \frac{-7}{8} \times \frac{10}{9}$

Cancel out common numerical factors across fractions.

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7 and 9 have a highest common factor of 1.
8 and 10 have a highest common facator of 2
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$= \frac{-7}{4} \times \frac{5}{9}$

Multiply numerator by numerator and denominator by denominator.

$= -\frac{35}{36}$

15. The product of two fractions is 1. If one of the fractions is $\frac{5x}{9y}$, what is the other fraction?

Let F be the other fraction.

Use the fact the product of $\frac{5x}{9y}$ and F is 1 to set up an equation.

$$\frac{5x}{9y} \times F = 1$$

Make F the subject of the equation.

Multiply both sides by the reciprocal of $\frac{5x}{9y}$, where the reciprocal can be found by swapping the numerator and denominator.

$$F = \frac{9y}{5x}$$

16. Simplify $p^5 \div p^3 \times p^4$.

Working from left to right, use the index law $A^m \div A^n = A^{m-n}$ to simplify $p^5 \div p^3$.

$= p^2 \times p^4$

Use the index law $A^m \times A^n = A^{m+n}$.

$= p^6$

17. Simplify $\frac{y^2 \times y^5}{y^3 \times y^7}$ ------

Use the index law $A^m \times A^n = A^{m+n}$ to simplify

= $\frac{y^{2+5}}{y^{3+7}}$

Evaluate the powers

= $\frac{y^7}{y^{10}}$

Use the index law $\frac{A^m}{A^n} = A^{m-n}$ to simplify

= y^{7-10}

Subtract the powers

= y^{-3}